Calculus

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As a physics student in high school, CS undergraduate in NTU, I never have a proper/formal introduction to calculus. This tutorial is my best effort to formalize calculus for real numbers and served as a cheat sheet as well

1 Derivative

Let $f \in \mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^n)$. In this tutorial, to simplify the analysis, we assume f being continuously differentiable.

We define the derivative $D : \mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^n) \to \mathbb{R}^m \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ [1] where

- $\mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^n)$: set of all continuously differentiable functions from $\mathbb{R}^m$ to $\mathbb{R}^n$
- $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$: set of all linear maps from $\mathbb{R}^m$ to $\mathbb{R}^n$

The derivative D is an operator acting on the set of all continuously differentiable functions from $\mathbb{R}^m$ to $\mathbb{R}^n$ and produce a linear map on every points on $\mathbb{R}^m$

The derivative of f at $x_0 \in \mathbb{R}^m$ is the *closest* linear map [2] to f at x_0

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o(x - x_0) \tag{1}$$

where $o(x - x_0) \subset \mathbb{R}^m \to \mathbb{R}^n$ is the set of functions strictly smaller than $||x - x_0||$ when x approaches x_0, i.e $\lim_{x \to x_0} \frac{||o(x-x_0)||}{||x-x_0||} = 0$

$Df(x)$ is a linear map from $\mathbb{R}^m$ to $\mathbb{R}^n$ which coincide with the Jacobian matrix $J_{f(x)} \in \mathbb{R}^{n \times m}$

$$J_{f(x)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_m} \\ \dots & \dots & \dots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_m} \end{bmatrix} \tag{2}$$

[1] **function currying notation**
$A \to B$: set of all functions from A to B
$A \to B \to C = A \to (B \to C)$

[2] the existence and uniqueness of $Df(x_0)$ is a corrolary of Taylor's Theorem

2 Derivative of element-wise function

Given a scalar function $\sigma \in \mathcal{C}^1(\mathbb{R}, \mathbb{R})$ with derivate $D\sigma(x) \in \mathcal{L}(\mathbb{R}, \mathbb{R})$ which is a 1d linear function, i.e $y = ax$ for $a, x \in \mathbb{R}$

We can generalize this function to n-dimensional $\bar{\sigma}(x) \in \mathcal{C}^1(\mathbb{R}^n, \mathbb{R}^n)$ such that if $y = \bar{\sigma}(x)$ for $x \in \mathbb{R}^n$, then we have

$$y_i = \sigma(x_i) \tag{3}$$

We can calculate the Jacobian matrix of this function

$$J = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & ... & \frac{\partial y_1}{\partial x_n} \\ ... & ... & ... \\ \frac{\partial y_n}{\partial x_1} & ... & \frac{\partial y_n}{\partial x_n} \end{bmatrix} \tag{4}$$

where

$$J_{i,j} = \begin{cases} D\sigma(x_i) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{5}$$

Jacobian matrix is diagonal

$$J = \begin{bmatrix} D\sigma(x_0) & ... & 0 \\ ... & ... & ... \\ 0 & ... & D\sigma(x_n) \end{bmatrix} \tag{6}$$

3 Chain Rule

Chain rule is used when we can to calculate the derivative of a function composition. Let's derive the chain rule.

Given $f \in \mathcal{C}^1(\mathbb{R}^n, \mathbb{R}^k)$, $g \in \mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^n)$, and $f \circ g \in \mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^k)$. We will calculate $D(f \circ g)$ given Df and Dg

Let $x, x_0 \in \mathbb{R}^m$ and $y = g(x), y_0 = g(x_0) \in \mathbb{R}^n$

$Df(y_0)$ is the derivative of f at y_0, we have

$$f(y) = f(y_0) + Df(y_0)(y - y_0) + o(y - y_0) \tag{7}$$

Rewrite

$$(f \circ g)(x) = (f \circ g)(x_0) + Df(g(x_0))(g(x) - g(x_0)) + o(g(x) - g(x_0)) \tag{8}$$

$Dg(x_0)$ is the derivative of f at x_0, we have

$$g(x) = g(x_0) + Dg(x_0)(x - x_0) + o(x - x_0) \tag{9}$$

Hence

$$
\begin{aligned}
(f \circ g)(x) &= (f \circ g)(x_0) + Df(g(x_0))(Dg(x_0)(x - x_0) + o(x - x_0)) + o(g(x) - g(x_0)) \\
&= (f \circ g)(x_0) + Df(g(x_0))Dg(x_0)(x - x_0) + Df(g(x_0))o(x - x_0) + o(g(x) - g(x_0))
\end{aligned}
$$

We can prove $Df(g(x_0))o(x-x_0)+o(g(x)-g(x_0)) = o(x-x_0)$ by assuming $||Df(g(x_0))|| < \infty$ and $||Dg(x_0)|| < \infty$

$Df(g(x_0))o(x - x_0)$ is small

$$
||Df(g(x_0))o(x - x_0)|| \leq ||Df(g(x_0))||||o(x - x_0)|| \qquad \text{(operator norm)}
$$

We have

$$
\begin{aligned}
\lim_{x \to x_0} \frac{||Df(g(x_0))o(x - x_0)||}{||x - x_0||} &\leq ||Df(g(x_0))|| \lim_{x \to x_0} \frac{||o(x - x_0)||}{||x - x_0||} \\
&= 0
\end{aligned}
$$

$o(g(x) - g(x_0))$ is small

$$
\begin{aligned}
\frac{||o(g(x) - g(x_0))||}{||x - x_0||} &= \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} \frac{||g(x) - g(x_0)||}{||x - x_0||} \\
&= \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} \frac{||Dg(x_0)(x - x_0) + o(x - x_0)||}{||x - x_0||} \\
&\leq \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} \frac{||Dg(x_0)(x - x_0)|| + ||o(x - x_0)||}{||x - x_0||} && \text{(Cauchy-Schwarz)} \\
&\leq \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} \frac{||Dg(x_0)||||(x - x_0)|| + ||o(x - x_0)||}{||x - x_0||} && \text{(operator norm)} \\
&= \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} (||Dg(x_0)|| + \frac{||o(x - x_0)||}{||x - x_0||})
\end{aligned}
$$

We have

$$
\begin{aligned}
\lim_{x \to x_0} \frac{||o(g(x) - g(x_0))||}{||x - x_0||} &\leq \lim_{x \to x_0} \frac{||o(g(x) - g(x_0))||}{||g(x) - g(x_0)||} (||Dg(x_0)|| + \frac{||o(x - x_0)||}{||x - x_0||} \\
&= 0
\end{aligned}
$$

Therefore,

$$
(f \circ g)(x) = (f \circ g)(x_0) + Df(g(x_0))Dg(x_0)(x - x_0) + o(x - x_0) \qquad (10)
$$

Theorem 1 (Chain Rule)**.** *Given two differentiable function* $f \in \mathcal{C}^1(\mathbb{R}^n, \mathbb{R}^k)$*,* $g \in \mathcal{C}^1(\mathbb{R}^m, \mathbb{R}^n)$*, the derivative of the composition* $f \circ g$ *is* $D(f \circ g)(x) = Df(g(x))Dg(x)$

4 Change of variables

Given a Lebesgue integral on a measurable subset $E \subset \mathbb{R}^n$ of positive measurable function $f : E \to [0, \infty)$

$$\int_E f \tag{11}$$

Let $G : \Omega \to E$ be a parameterization satisfying certain conditions [3] where Ω. Then

$$\int_E f = \int_\Omega (f \circ G)|DG| \tag{12}$$

where $|DG|$ is the determinant of derivative of G

References

[3] $\mathcal{C}^1$ diffeomorphism